

```
!pip install yellowbrick
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
!wget https://archive.ics.uci.edu/ml/machine-learning-databases/letter-recognition/letter-recognition.data
```

```
--2026-04-01 19:10:36-- https://archive.ics.uci.edu/ml/machine-learning-databases/letter-recognition/letter-recognition.dat
Resolving archive.ics.uci.edu (archive.ics.uci.edu)... 128.195.10.252
Connecting to archive.ics.uci.edu (archive.ics.uci.edu)|128.195.10.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified
Saving to: 'letter-recognition.data.1'

letter-recognition.      [ <=>          ] 695.86K  2.12MB/s   in 0.3s

2026-04-01 19:10:36 (2.12 MB/s) - 'letter-recognition.data.1' saved [712565]
```

```
X = dataset.iloc[:, 1:17]
Y = dataset.iloc[:, 0].values.ravel() # FIXED (important)
```

```
X_train, X_validation, Y_train, Y_validation = train_test_split(
    X, Y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_validation = scaler.transform(X_validation)
```

```
mlp = MLPClassifier(
    hidden_layer_sizes=(100, 100),
    max_iter=300,
    activation='relu',
    solver='adam',
    random_state=42
)
```

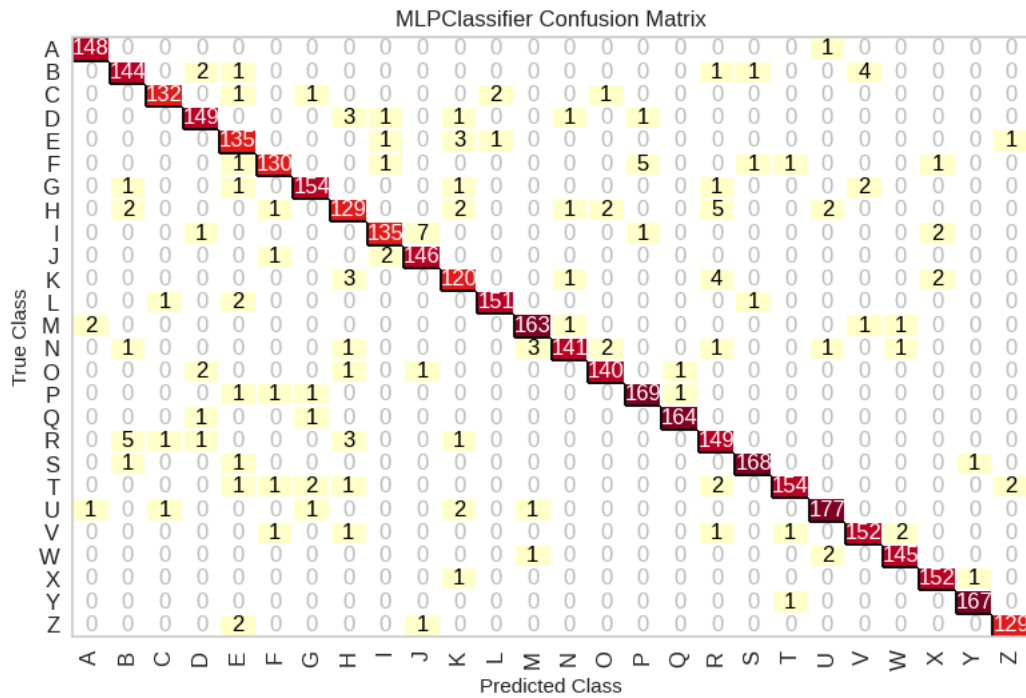
```
mlp.fit(X_train, Y_train)
```

```
MLPClassifier
MLPClassifier(hidden_layer_sizes=(100, 100), max_iter=300, random_state=42)
```

```
from yellowbrick.classifier import ConfusionMatrix

classes = list("ABCDEFGHIJKLMNOPQRSTUVWXYZ")

cm = ConfusionMatrix(mlp, classes=classes)
cm.fit(X_train, Y_train)
cm.score(X_validation, Y_validation)
cm.show()
```



```
predictions = mlp.predict(X_validation)
print(predictions)
```

```
['T' 'L' 'A' ... 'Q' 'Y' 'Y']
```

```
print("Accuracy:", accuracy_score(Y_validation, predictions))
```

```
Accuracy: 0.96075
```

Start coding or [generate](#) with AI.